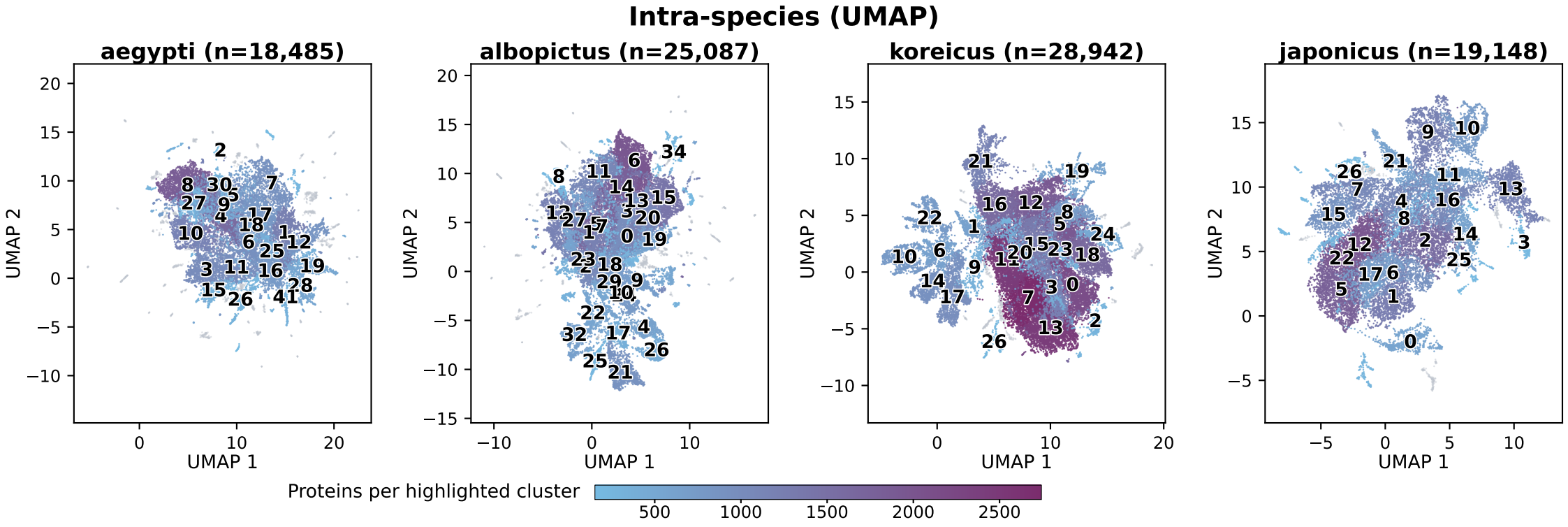
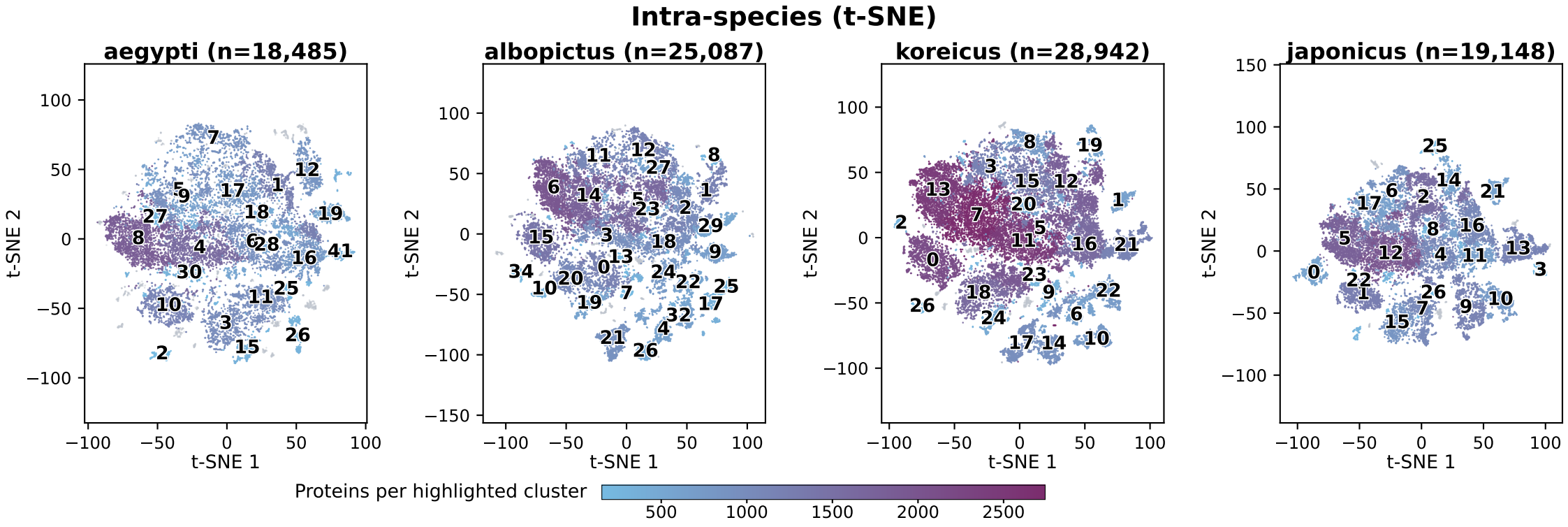
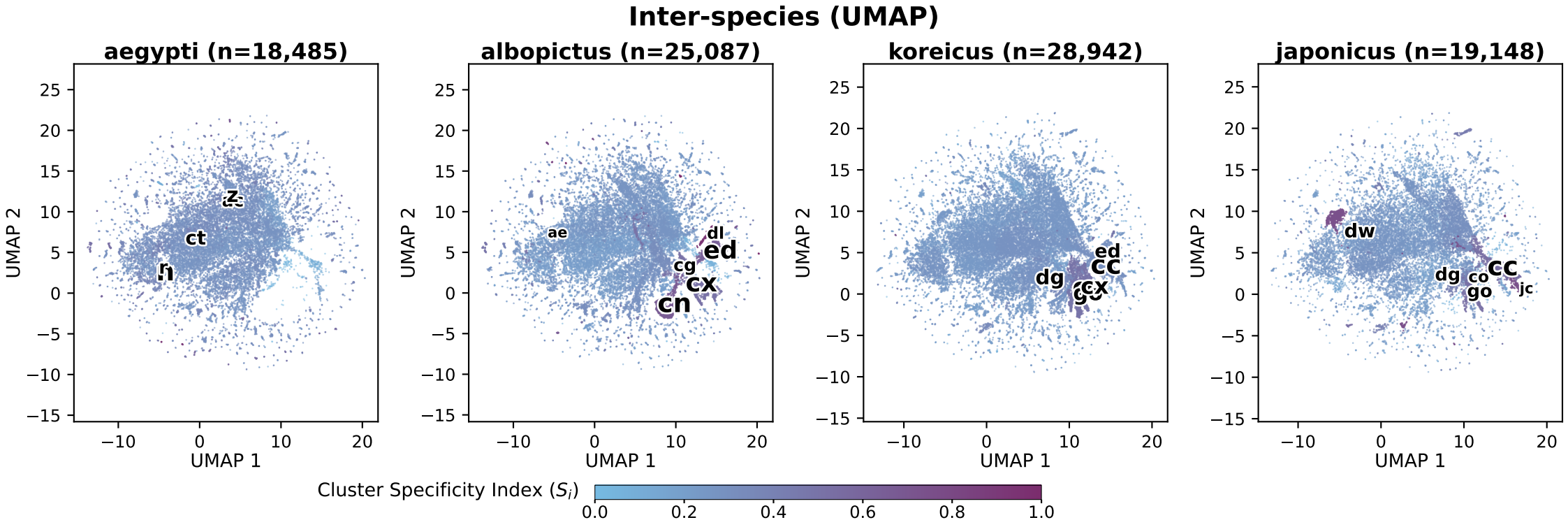
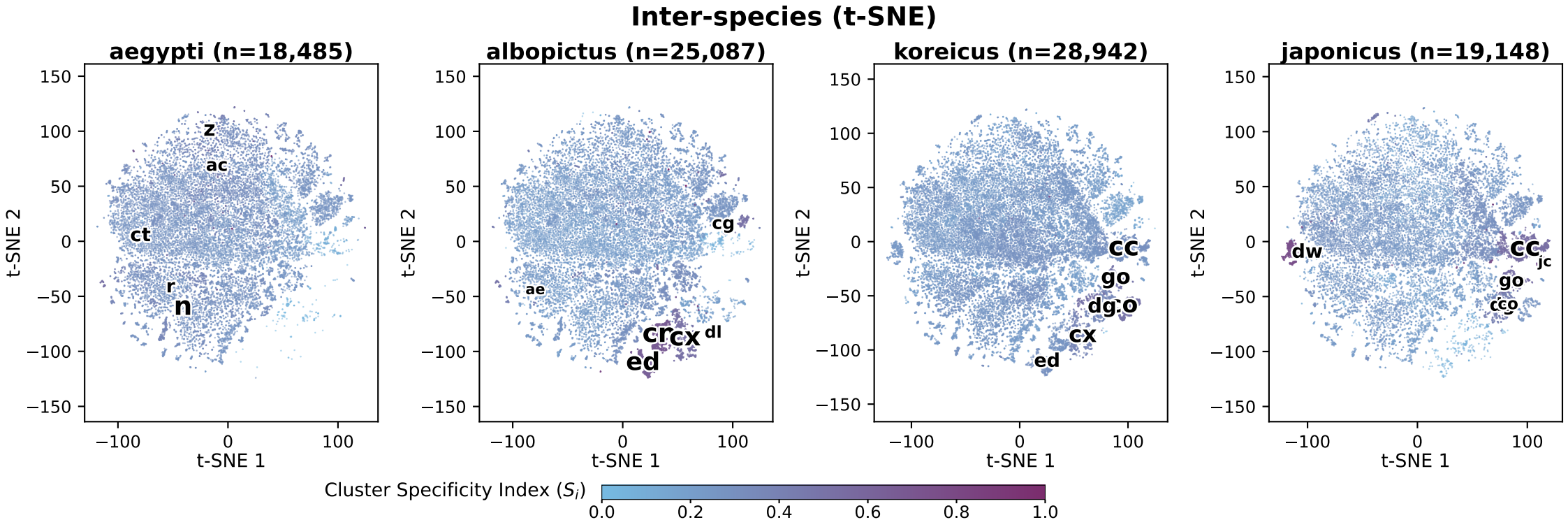
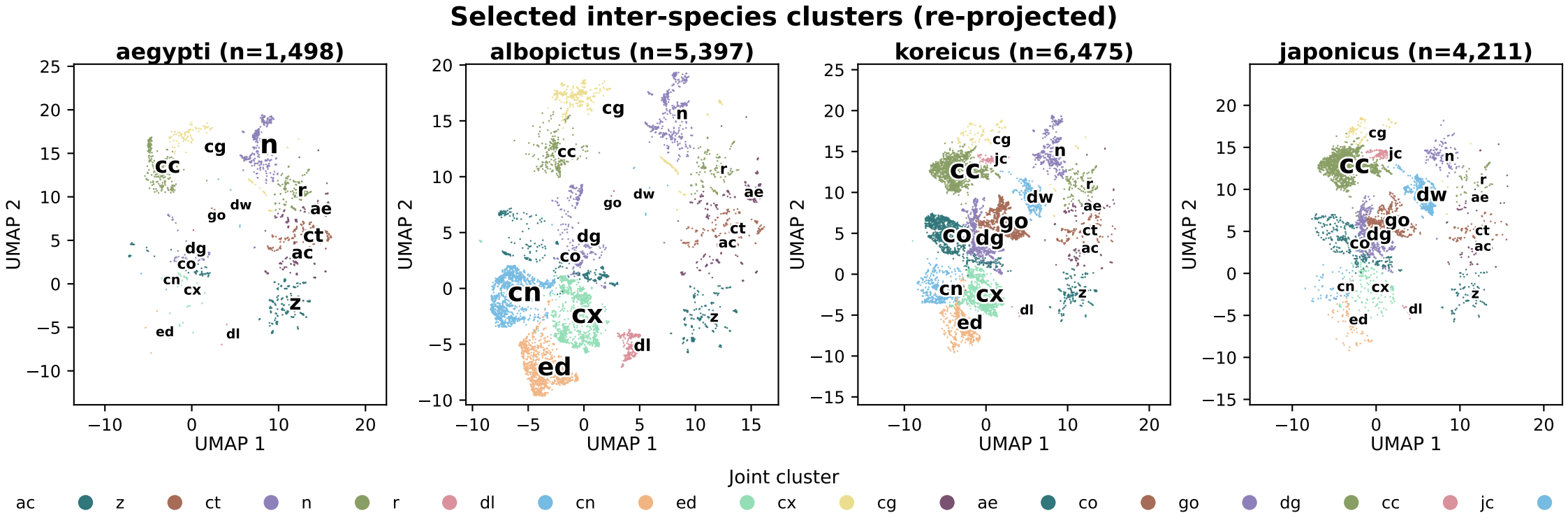

Intra-species (UMAP)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
UMAP 1
UMAP 2
Proteins per highlighted cluster
500
1000
1500
2000
2500


Intra-species (t-SNE)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
t-SNE 1
t-SNE 2
Proteins per highlighted cluster
500
1000
1500
2000
2500


Inter-species (UMAP)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
UMAP 1
UMAP 2
ct
ae
dl
ed
cg
cx
cn
dg
cc
co
go
jc
dw
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Inter-species (t-SNE)
aegypti (n=18,485)
albopictus (n=25,087)
koreicus (n=28,942)
japonicus (n=19,148)
t-SNE 1
t-SNE 2
z
ac
ct
r
n
cg
ae
cr
cx
dl
ed
cc
go
dg
o
cx
ed
dw
cc
jc
go
cc
o
Cluster Specificity Index (S_i)
0.0
0.2
0.4
0.6
0.8
1.0


Selected inter-species clusters (re-projected)
aegypti (n=1,498)
albopictus (n=5,397)
koreicus (n=6,475)
japonicus (n=4,211)
UMAP 1
UMAP 2
Joint cluster
ac
z
ct
n
r
dl
cn
ed
cx
cg
ae
co
go
dg
cc
jc

Intra- to Inter-species cluster fluxes

aegypti (n=18,485, k=70)

8
4
1
10
3
11
5
6
7
17
16
9
26

ac (110)
z (198)
ct (159)
n (335)
r (138)

albopictus (n=25,087, k=77)

5
15
1
2
18
21
24
32
4
22
29
26
17
25

dl (263)
cn (1,028)
ed (906)
cx (990)
cg (355)
ae (119)

koreicus (n=28,942, k=42)

7
13
11
16
21
17
14
10
6
9

co (858)
go (696)
cx (829)
dg (624)
cc (1,118)
ed (493)

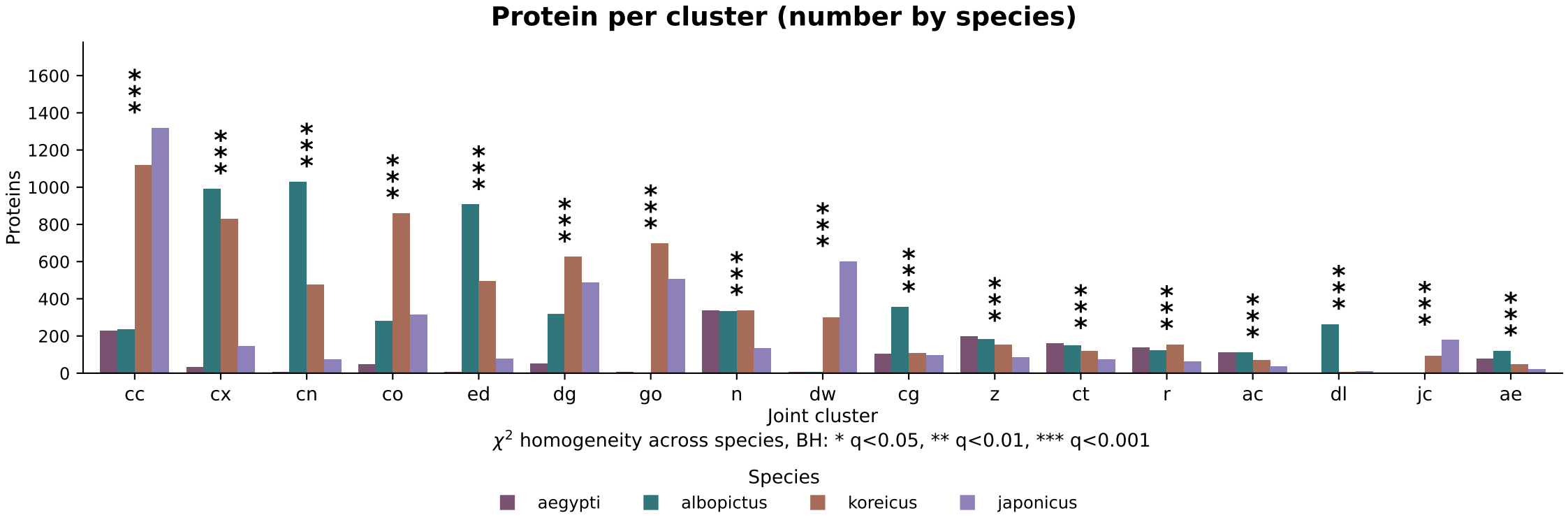
japonicus (n=19,148, k=30)

5
9
13
16
10
11
14
0
3

jc (177)
dw (600)
cc (1,318)
go (506)
dg (487)
co (313)

Cluster Specificity Index (S_i); black border: $S_i \geq 0.80$

0.0 0.2 0.4 0.6 0.8 1.0


Protein per cluster (number by species)
Proteins
1600
1400
1200
1000
800
600
400
200
0
***
cc
cx
cn
co
ed
dg
go
n
dw
cg
z
ct
r
ac
dl
jc
ae
Joint cluster
χ^2 homogeneity across species, BH: * q<0.05, ** q<0.01, *** q<0.001
Species
aegypti
albopictus
koreicus
japonicus